MIL-W-26497C
9 November 1977
SUPERSEDING
MIL-W-26497B
6 March 1969

MILITARY SPECIFICATION

WRENCH, TORQUE, SCREWDRIVER TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of screwdriver type torque wrenches.

1.2 Classification. Torque wrenches shall be of the following types and capacities as specified (see 6.2 and 6.3).

Type I - Adjustable

Class A - 1/4 inch (6.3 mm) hexagon socket drive
Class B - 1/4 inch (6.3 mm) square male drive

Capacity 1 - 1-20 ounce-inch (.007- .141 N·m)
Capacity 2 - 1-100 ounce-inch (.007- .706 N·m)
Capacity 3 - 1.30 pound-inch (.113-3.39 N·m)
Capacity 4 - 2-36 pound-inch (.226-4.07 N·m)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5120

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Type II - Direct reading

Class A - 1/4 (6.3 mm) hexagon socket drive
Class B - 1/4 (6.3 mm) square male drive

Capacity 1 - 0-12 ounce-inch (0-.084 N·m)
Capacity 2 - 0-6 pound-inch (0-.680 N·m)
Capacity 3 - 0-25 pound-inch (0-2.82 N·m)
Capacity 4 - 0-50 pound-inch (0-5.64 N·m)
Capacity 5 - 0-100 pound-inch (0-11.3 N·m)

Style a - Right and left hand indicating
Style b - Right hand indicating
Style c - Left hand indicating

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GGG-B-1222 (GSA-FSS) - Bit, Screwdriver (and Adapter, Screwdriver Bit)
PPP-P-40 - Packaging and Packing of Hand Tools

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Standard product. Except for differences that are specified herein, the torque wrenches delivered under requirements of this specification shall be a model and configuration of an item that is available on a commercial basis.

3.2 Materials (see 6.3). Materials not definitely specified shall be of the quality conforming to the manufacturer's current commercial practice.

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3.3 Design and construction.

3.3.1 Type I, adjustable torque. The adjustable screwdriver type torque wrench shall consist of a handle with a socket and a clutch type device that automatically releases when a preset torque is reached. The torque wrench shall indicate torque in both right hand (clockwise) and a left hand (counter-clockwise) direction. The torque wrench shall be adjustable throughout the entire range. Torque selection shall be accomplished by presetting on a graduated dial to the required torque. When the preset torque value is reached during operation of the wrench, the handle shall go into a freewheeling condition with no further turning of the threaded fastener possible. The type I, class A torque wrench shall be a 1/4 inch (6.3 mm) hexagon socket drive integral with the wrench. The type I, class B torque wrench shall have a 1/4 inch (6.3 mm) square male drive integral with the wrench. The capacity and overall length shall be as specified in table I.

TABLE I. Type I torque wrenches

Capacity	Graduations	Overall length (minimum)
1	1/4 ounce-inch (.002 N·m)	3-7/8 inches (98.4 mm)
2	2 ounce-inch (.014 N·m)	6-3/8 inches (162 mm)
3	1 pound-inch (.113 N·m)	6-1/2 inches (165 mm)
4	2 pound-inch (.226 N·m)	7-1/2 inches (190 mm)

3.3.1.1 Adjustment mechanism. The torque adjustment mechanism shall consist of a barrel-type scale and shall be clearly visible for adjusting or setting. The required torque shall be set by adjusting the scale to the required setting without the use of an adjusting tool. The control mechanism shall engage and disengage a minimum of four times in one complete revolution when tested as specified in 4.3.1.1. The wrench shall be so designed that it cannot be overtorqued beyond its design range.

3.3.1.2 Scale graduations. The scale graduations shall be permanent, well defined, and clearly visible. The torque wrenches shall be graduated as shown in table I.

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3.3.2 Type II, direct reading. The direct reading, screwdriver type torque wrench shall consist of a handle with a round dial and an indicator or a deflecting beam indicator attached to the base of the handgrip to visually indicate torque. Alternatively, a deflecting beam type with indicator plate may be furnished. The capacity, overall length and graduations shall be as specified in table II. The type II, class A torque wrench shall have a 1/4 inch (6.3 mm) hexagon socket drive integral with the wrench to accomodate the screwdriver bits (see 4.20 MIL-STD-961). The type II, class B torque wrench shall have a 1/4 inch (6.3 mm) square male drive integral with the wrench. The type II, capacities 1 and 2 torque wrenches shall be furnished in style a, right and left hand indicating. The type II, capacities 3, 4 and 5 shall be furnished in styles b or c, right and left hand indicating, as required.

TABLE II. Type II torque wrenches

Capacity	Graduations	Overall length (minimum)
1	1 ounce-inch (.007 N·m)	8-1/2 inches (216 mm)
2	1/2 pound-inch (.056 N·m)	8-9/16 inches (217 mm)
3	1 pound-inch (.113 N·m)	11-1/2 inches (292 mm)
4	2 pound-inch (.226 N·m)	11-3/8 inches (289 mm)
5	5 pound-inch (.565 N·m)	12-3/8 inches (314 mm)

3.3.2.1 Screwdriver bits and square to hexagon drive adapters. The bits shall conform to the requirements for type II bits specified in GGG-B-1222 and shall be furnished in the type and sizes specified (see 6.2). The adapter shall be 1/4 inch (6.3 mm) square to 1/4 inch (6.3 mm) hexagon male to male or female to female, as specified (see 6.2). The adapters shall be no less than 1 inch (25.4 mm) nor more than 2 inches (50.8 mm) overall length.

3.3.2.2 Handgrip. The handgrip for the type II wrench shall be fabricated from plastic and shall be fluted. The hand grips shall be capable of withstanding the flammability and shatter tests specified in 4.2.1.1.

3.4 Performance.

3.4.1 Control and accuracy.

3.4.1.1 Type I. The type I wrench shall be capable of controlling the applied torque in either a left hand or right hand direction. The wrench shall control the applied torque within the tolerances and at the percent capacities specified in table III. There shall be no tendency to back off the tightened fastener once the required torque delivery has been achieved. The wrenches shall conform to these requirements when tested as specified in 4.3.2 and 4.3.3.

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3.4.1.2 Type II. The type II wrench shall be capable of controlling the applied torque in a right hand, left hand, or both directions, as applicable, within the tolerances, and at the capacities specified in table III. When the applied torque is released, the indicator shall return to the zero position after testing as specified in 4.3.2 and 4.3.3.

TABLE III. Wrench tolerances and capacities

WRENCH CAPACITY	APPLIED TORQUE			
	25% of wrench capacity	50% of wrench capacity	75% of wrench capacity	100% of wrench capacity
Type I:				
Capacity 1	1/4 ounce-inch (.002 N·m)	3/8 ounce-inch (.003 N·m)	1/2 ounce-inch (.004 N·m)	3/4 ounce-inch (.006 N·m)
Capacity 2	2 ounce-inch (.014 N·m)	3 ounce-inch (.021 N·m)	4 ounce-inch (.028 N·m)	5 ounce-inch (.035 N·m)
Capacity 3	3/4 pound-inch (.084 N·m)	1 pound-inch (.113 N·m)	1.25 pound-inch (.141 N·m)	1-1/2 pound-inch (.156 N·m)
Capacity 4	1 pound-inch (.113 N·m)	1-1/2 pound-inch (.156 N·m)	2 pound-inch (.226 N·m)	2-1/2 pound-inch (.282 N·m)
Type II:				
Capacity 1	1/2 ounce-inch (.004 N·m)	1/2 ounce-inch (.004 N·m)	3/4 ounce-inch (.006 N·m)	1 ounce-inch (.007 N·m)
Capacity 2	1/4 pound-inch (.028 N·m)	3/8 pound-inch (.042 N·m)	1/2 pound-inch (.056 N·m)	3/4 pound-inch (.084 N·m)
Capacity 3	3/4 pound-inch (.084 N·m)	1 pound-inch (.113 N·m)	1-1/4 pound-inch (.141 N·m)	1-1/2 pound-inch (.156 N·m)
Capacity 4	1 pound-inch (.113 N·m)	1-3/4 pound-inch (.198 N·m)	2 pound-inch (.226 N·m)	2-1/2 pound-inch (.282 N·m)
Capacity 5	2-1/2 pound-inch (.282 N·m)	3 pound-inch (.339 N·m)	4 pound-inch (.452 N·m)	5 pound-inch (.565 N·m)

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3.4.2 Overload. When tested as specified in 4.3.4, the torque wrenches shall be capable of withstanding a 25 percent overload in either rotational direction without any deformation when the wrench measures left and right, and in the direction of the reading when it is a one directional wrench.

3.4.3 Dropping. The torque wrenches shall not become inoperable and deformed when tested as specified in 4.3.5, except that the dial for the type II may be deformed but shall remain operational.

3.5 Marking for identification. The torque wrench shall be permanently marked with the manufacturer's model number and name or readily identifiable trademark.

3.6 Finish. All exterior metal parts, except nonferrous or corrosion-resistant metal, shall be plated or coated in accordance with the manufacturer's current commercial practice, except that the use of cadmium plating shall not be allowed.

3.7 Workmanship. The wrenches shall be free of sharp edges, splinters, burrs or chips. The assembly screws shall be tight and all components shall be aligned. There shall be no indication of lack of plating or coating, foreign matter imbedded, or blisters on plated or coated surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.2.1.1 In-process testing of type II handle. The handle shall be tested for shatter and flammability resistance as specified in 4.3.6. The lot size shall be all the handles furnished for inspection at one time. The inspection level shall be S-1. Any evidence of failure of these tests shall be cause for rejection of the lot.

4.2.2 End item inspection. The inspection lot shall consist of all torque wrenches of the same type and capacity offered for inspection at one time. The sample unit shall be one completely fabricated torque wrench.

4.2.2.1 Visual examination. The completely assembled torque wrench shall be examined for defects in table IV. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE IV. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not plated or coated where required	X	
	Signs of rust or corrosion	X	
	Foreign matter imbedded or blisters		X
	Cadmium plating used	X	
Construction and workmanship	Type and range not as specified	X	
	Sharp edges, splinters, burrs or chips	X	
	Markings and graduations not as specified	X	
	Not evenly or not clearly marked	X	
	Pointer bent, malformed or not readily visible (type II)	X	
	Not 1/4 inch (6.3 mm) drive	X	
	Screwdriver bits not retained securely by socket (when applicable)	X	
	Barrel-type scale (type I) not as specified, not clearly visible	X	
	Torque setting mechanism cannot be readily adjustable (type I)	X	
	Drive adapter missing (when applicable)	X	
Marking for identification	Illegible, incorrect, or missing		X

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4.2.2.2 Dimensional examination. Inspection shall be made of the torque wrench for compliance with dimensional requirements. Any deviation from the specific requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects, per hundred units.

4.2.2.3 End item testing. The type I and II torque wrenches shall undergo the applicable tests specified in 4.3. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.4 Packaging inspection. The sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with the requirements of PPP-P-40.

4.3 Tests.

4.3.1 Torque test.

4.3.1.1 Torque test, type I. The shank shall be held firmly and the torque wrench rotated around the shank centerline in both a clockwise and counterclockwise direction. When the wrench is applied to a known torque, the torque wrench shall indicate the specified torque in each direction, when applicable, in one complete revolution of the torque wrench. The test shall be repeated five times. Any evidence of non-conformance with the requirements specified in 3.3.1.1 shall constitute failure of the test.

4.3.1.2 Torque test, type II. When the type II torque wrench is applied to a known torque, the torque wrench shall indicate the specified torque and return to the zero position when released. Those wrenches that indicate in both clockwise and counterclockwise direction shall be tested in both directions. The test shall be repeated five times. Any evidence of non-conformance with the requirements specified in 3.4.1.2 shall constitute failure of the test.

4.3.2 Accuracy test. The torque wrench shall be tested for accuracy by the dead weight test or equivalent test at room temperature. The torque wrench shall be tested at 25, 50, 75 and 100 percent of capacity of the wrench scale. The test shall be repeated 5 times. If the wrench indicates in both clockwise and counterclockwise directions, the complete test shall be performed in each direction. Any evidence of nonconformance with the requirements specified in 3.4.1.1 or 3.4.1.2, as applicable, shall constitute failure of the test.

4.3.3 Cycling test. The torque wrench shall be subjected to 1000 cycles at a rate of not less than 6 cycles per minute (cpm) at each setting of 25, 50, 75 and 100 percent of capacity at room temperature. At the completion of each

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test, the torque wrench shall be examined to determine compliance with the accuracy and sensitivity requirements specified in 3.4.1.1 or 3.4.1.2, as applicable. Any evidence of nonconformance shall constitute failure of the test.

4.3.4 Overload test. The torque wrench shall be subjected to 25 percent overload of the maximum capacity in the clockwise and counterclockwise directions. At the conclusion of this test, the torque wrench shall be re-subjected to the test specified in 4.3.3 and shall maintain the accuracy specified. The torque wrench shall then be examined for compliance with 3.4.2. For the type I wrench, the stops on the adjustment mechanism shall be altered to enable this test to be performed. Any evidence of nonconformance shall constitute failure of the test.

4.3.5 Drop test. The wrench shall be dropped a distance of 5 feet (1.53 m) onto a concrete slab three times; first, the wrench shall be dropped onto one end, then dropped to land on the other end and finally dropped to land on one of the sides. The wrench shall then be rechecked for accuracy by the test specified in 4.3.3. Any evidence of nonconformance shall constitute failure of the test.

4.3.6 Plastic handgrip test (type II).

4.3.6.1 Shatter test. Handgrip shall be placed on a solid foundation and a suitable guard shall be placed around the handgrip. A weight of approximately 5 pounds (2.25 kg) shall be dropped directly upon the handgrip from a 3-foot (0.9 m) height. Test shall be repeated. If the handgrip is not still intact after the two direct blows, it shall constitute failure of the shatter test.

4.3.6.2 Flammability test. Handgrip shall be clamped in a vise and a flame from blowtorch, Bunsen burner, or equivalent shall be applied to one end for 30 seconds. If this does not ignite the material, flame shall be applied the second time for 30 seconds. If the second application ignites the handgrip and the rate of burn exceeds 2 inches (50.8 mm) per minute it shall be considered fast burning and shall constitute failure of the test.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging, packing and marking for level A, B or C shall be in accordance with PPP-P-40.

6. NOTES

6.1 Intended use. The screwdriver type torque wrenches are used to apply to the proper torque required for the installation and removal of threaded fasteners.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, capacity and style (if applicable) of torque wrench.
(c) When bits are required (see 3.3.2.1).
(d) When adapter is required (see 3.3.2.1).
(e) Selection of applicable levels of preservation-packaging and packing (see 5.1).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

6.4 Supersession data.

"B" revision	"C" revision
Type I - Adjustable-Magnetic	Type I - Adjustable
4 - 100 ounce-inch, 4 ounce-inch, 7 inch long	1 - 100 ounce-inch, 2 ounce-inch, 6-3/8 inch long

6.5 Changes from the previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - AS
Air Force - 99

Review activity:

Army - AV

Preparing activity:

Army - GL

Project No. 5120-0725

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-W-26497C; Wrench, Torque, Screwdriver Type

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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